

PARTS LIST

Semiconductors:

- IC1 - ADE7757 energy metering IC
- IC2 - AT89C52 microcontroller
- IC3 - AT24C02 EEPROM
- IC4 - MCT2E opto-coupler
- IC5, IC6 - 7805, 5V regulator
- T1 - BC547 npn transistor
- D1 - 1N4007 rectifier diode
- LED1, LED2 - 5mm LED
- LC01 - 16x2 LCD
- BR1 - Bridge rectifier module 1A
- ZD1 - 15V, 1W zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R3, R7, R8 - 499-ohm
- R2 - 6.2-kilo-ohm
- R4 - 1-kilo-ohm
- R5, R6, R19 - 680-ohm
- R9 - 350-micro-ohm shunt
- R10 - 1-mega-ohm
- R11 - 470-ohm, 1W
- R12-R16 - 10-kilo-ohm
- R17 - 100-ohm
- R18 - 470-ohm
- VR1 - 500-kilo-ohm preset
- VR2 - 10-kilo-ohm preset

Capacitors:

- C1, C6, C16 - 10 μ F, 16V electrolytic
- C2, C3, C7-C10, C17, C21 - 0.1 μ F ceramic disk
- C4, C5, C11, C12 - 0.068 μ F ceramic disk
- C13 - 0.47 μ F, 630V polyester
- C14 - 470 μ F, 35V electrolytic
- C15 - 0.01 μ F ceramic disk
- C18, C19 - 22pF ceramic disk
- C20 - 1000 μ F, 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- X₂, X₃ - 1MOV
- L1, L2 - Ferrite bead
- L3, L4 - 3.5mm $\times$ 9mm axial, filter choke (bead core)
- SI - Tactile switch
- CON1, CON2 - 2-pin 5mm terminal connector
- CON3 - 2-pin connector

opto-coupler MCT2E (IC4) and an LCD display.

IC ADE7757. It is a low-cost, single-chip solution for electrical energy measurement. In operation, the chip interfaces with a shunt resistor (used as the current sensor) and AC analogue voltage sensing the inputs and outputting the consumed power as explained below. It has two analogue input channels designated as V1 and V2, respectively. Channel V1 (also called 'current channel') is used for current sensing and channel V2 (also called 'voltage channel') is used for voltage sensing. The differential output from the current-sensing resistor is connected between V1P and V1N inputs, whilst the differential output signal proportional to the AC line voltage,

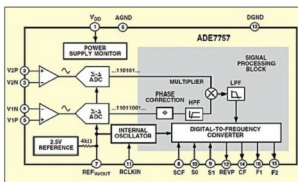


Fig. 2: Functional block diagram of ADE7757

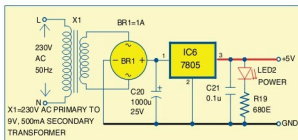


Fig. 3: Additional power supply for the counter circuit

TABLE II
Output Frequency on CF Pin

SCF	S1	S0	CF signals (Hz)
1	0	0	128 X F1, F2
0	0	0	64 X F1, F2
1	0	1	64 X F1, F2
0	0	1	32 X F1, F2
1	1	0	32 X F1, F2
0	1	0	32 X F1, F2
1	1	1	16 X F1, F2
0	1	1	2048 X F1, F2

obtained through a resistor divider, is connected between pins V2P and V2N.

IC ADE7757 has a reference circuit and a fixed DSP function for calculation of the real power. A highly stable oscillator integrated into the chip provides the necessary clock for the chip. It supplies the average real-power information on the F1 and F2 low-frequency outputs. The meter is designed for 100 pulses/kWh and the pulses can be counted by any counter for power consumption calculation. Here, microcontroller AT89C52 is used for counting the pulses. ADE7757 provides a high-frequency output at the calibration frequency (CF) pin (here, it is 3200 pulses/kWh), which is selected via S1

and S0 pins as shown in bold in Table II. This high-frequency output provides instantaneous real-power information, which is used to speed up the calibration process. Functional block diagram of ADE7757 is shown in Fig. 2.

The power supply for IC ADE7757 is derived directly from mains using the capacitor divider network comprising C13 and C14. Most of the voltage is dropped across C13 (0.47 μ F polyester capacitor rated for 630V), whilst resistor R11 (470-ohm, 1W) is used as the current limiter. The output across

C14 is limited to 15V DC, which serves as an input to regulator IC5. The regulated 5V is fed to IC1. The F1 output of IC1 is coupled to port pin P3.2 of microcontroller IC2 via opto-coupler IC4 whilst LED1 indicates that IC1 is working.

AT89C52. It is a low-power, high-performance CMOS 8-bit microcontroller that provides standard features: 8k bytes of Flash, 256 bytes of RAM, 32 I/O lines, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

Microcontroller IC2 takes the meter reading through its pin 12 and stores it in EEPROM IC3, and at the same time displays it on the LCD, which requires additional 5V regulated and isolated supply (to avoid extension of live mains to the counter section). A conventional 5V regulator circuit incorporating a bridge rectifier (BR1), smoothing capacitor (C20) and regulator IC 7805 (IC6) has been used for the purpose. Fig. 3 shows the additional power supply.

Pins 21 through 28 of microcontroller IC2 are connected to the LCD data pins D0 through D7, respectively.